INCH-POUND

MIL-R-82784(OS)
5 February 1990
SUPERSEDING
WS 22372A
9 December 1986

MILITARY SPECIFICATION

ROD, PLASTIC

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers one type of polyester plastic rod, referred to herein as the plastic rod.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-313 *Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities*

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:25Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 **Other Government documents, drawings and publications.** The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those listed in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 2101 Tensile Properties of Single Man Made Textile Fibers Taken from Yarns and Tows

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:25Z
Check the source to verify that this is the current version before use.

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample of the plastic rod shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 **Material requirements.** The plastic rod furnished in accordance with this specification shall be an extruded polyethylene terephthalate plastic material.

3.3 **Physical properties.** The physical properties of the plastic rod shall conform to table I.

TABLE I. Physical properties.

Property	Unit	Limits	
		Minimum	Maximum
Diameter	Inch	0.019	0.021
Tensile strength at break	Pound	13	-----
Shrinkage at 392 ± 5 degrees Fahrenheit (°F)	Percent	-----	9

3.4 **Safety**

3.4.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.4.2 **Material Safety Data Sheet (MSDS).** The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

3.5 **Shelf life.** The plastic rod shall meet the requirements of this specification for an unlimited period from the date of the quality conformance inspection (see 4.4) when packaged as specified in section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:25Z
Check the source to verify that this is the current version before use.

3.6 Workmanship. Workmanship shall be such that plastic rod is uniform in appearance and free from visible contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in he specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Toxicological product formulations. The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection (see 3.1) shall consist of the examination and tests specified in table II. Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:25Z
Check the source to verify that this is the current version before use.

TABLE II. Quality conformance inspection.

Examination and Tests	Requirement Paragraph	Inspection Paragraph
Visual examination	3.6 and section 5	4.5.1
Diameter	3.3	4.5.2
Shrinkage	3.3	4.5.3
Tensile strength	3.3	4.5.4
Packaging inspection	Section 5	4.6

4.4.1 Sampling for inspection.

4.4.1.1 Lot. For inspection purposes, a lot shall consist of all the material offered for acceptance at one time from a continuous process (see 6.5.1).

4.4.1.2 Selection of sample containers. Sample containers of the plastic rod shall be selected at random from each lot in accordance with table III.

TABLE III. Sampling for inspection.

Number of Containers in Lot	Number of Containers to be Selected for Sampling
1 - 3	ALL
4 - 15	3
16 - 25	4
26 - 40	5
41 - 65	7
66 - 110	10
111 - 180	15

4.4.1.3 Sample material. Sufficient material shall be selected at random from each sample container to perform all of the inspections specified in 4.5.1 through 4.5.4.

4.4.1.4 Noncompliance of samples. Failure of one or more of the sample containers of material to meet all of the specified requirements shall be cause for rejection of the lot.

4.5 Methods of inspection. The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes,

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:25Z
Check the source to verify that this is the current version before use.

temperatures, and times shall be measured to the nearest specified unit or decimal.

4.5.1 **Visual examination.** All samples shall be visually examined for conformance to 3.6.

4.5.2 **Diameter.** The diameter of the plastic rod shall be determined as follows:

a. Measure and cut approximately one meter of sample rod.

b. Using a micrometer, measure the diameter at three equally-spaced locations.

c. Record the individual measurements and the average diameter of the plastic rod.

4.5.3 **Shrinkage.** Shrinkage shall be determined in accordance with the following:

a. Measure the specimen (from 4.5.2) to the nearest millimeter (mm), (L1).

b. Preheat the oven to 392 $\pm$ 5°F.

c. Coil the plastic rod and hang on a hook in the oven.

d. Maintain oven temperature for 10 $\pm$ 1 minutes.

e. Remove specimen from the oven and cool to ambient room temperature.

f. Measure the sample to the nearest mm, (L2).

g. Calculate the shrinkage as follows:

$$\text{Shrinkage, percent} = \frac{L1 - L2}{L1} 100$$

Where: L1 = Original length of sample, mm

L2 = Shrinkage length, mm

h. Record the shrinkage to the nearest 0.1 percent.

4.5.4 **Tensile breaking strength.** The tensile breaking strength shall be determined in accordance with ASTM D 2101.

4.6 **Inspection of packaging.** The plastic rod containers and container markings shall be visually examined to verify compliance with section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:25Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 Level C. The plastic rod shall be packaged to afford adequate protection against loss, contamination, deterioration and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period specified in 3.5. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to them mode of transportation.

5.2 Marking.

5.2.1 Standard marking. In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 Special marking. In addition to the marking requirements of 5.2.1, each container marking shall include the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

c. Material trade name and identification.

d. Lot number and date of quality conformance inspection (see 4.4).

e. Toxicity warning, if applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but in not mandatory.)

6.1 Intended use. The plastic rod specified herein is intended for use in the manufacture of Standard Missile MK 104 Dual Thrust Rocket Motor igniters.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2, and 2.2).

c. Whether first article is required (see 3.1 and 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:25Z
Check the source to verify that this is the current version before use.

d. Applicable contract provisions for Material Safety Data Sheet (see 3.4.2 and 6.4).

e. Packaging required, if different than 5.1.1.

6.3 **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors on the material to be inspected. The material should be a first article sample and should consist of a quantity of the plastic rods sufficient to perform all of the inspections specified in 4.3. the first article sample should be packaged and marked as specified in section 5. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 **Material Safety Data Sheet (MSDS).** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR), SUBPART 23.3. Contracting officers will identify those activities requiring copies of completed prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 **Definition.**

6.5.1 **Continuous process.** A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the material is accumulated for a lot.

6.6 **Possible material.** Typical material defined by this specification is Polyester D9 monofilament manufactured by Albany International, Homer, N.Y.

6.7 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 1340-N792)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:25Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-82784	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
ROD, PLASTIC

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:25Z
Check the source to verify that this is the current version before use.